Gating AI Deployment on Environmental Evidence Quality

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Abstract

Carbon-aware governance proposals gate AI deployment on impact magnitude: a carbon budget, a readiness level, or a risk rating. None of them asks whether the carbon number itself can be trusted. We present EVIDENCEGATE, an environmental governance module deployed inside an open-source AI governance platform, that makes measurement provenance a first-class release criterion. Each assessment maps its measurement source to one of five provenance classes, which cap a numeric evidence-confidence score (vendor-reported figures cap at 0.60; undisclosed provenance forces 0.0); confidence crossed with an impact tier in a 3×3 decision matrix yields *go*, *conditional_go*, or *no_go*, and the recommendation blocks the platform's deployment-approval workflow. Offsets and renewable certificates never improve confidence, tier, or recommendation. On a 14-case hand-labeled fixture set the full mechanism reproduces all expected decisions, while ablated gates that see only carbon magnitude agree with at most half of them: a consistency check, not a field result, which is why we also present the design of two registered studies now instrumented and ready to run. A measurement study measures identical workloads three ways, so the same true impact carries different confidence and can flip the gate; a two-arm expert panel tests whether the matrix changes reviewer coverage and agreement. We seek workshop feedback on both designs before data collection completes.

1 The gap: a measurement layer with no trust layer

Regulators and standards bodies are building the measurement layer for AI's environmental footprint: quantification-methods inventories [1], sustainability overviews [2], measurement frameworks in progress, and a G7-commissioned inventory cataloguing thirteen measurement, monitoring, and reporting initiatives [3]. The EU AI Act anticipates harmonised standards for "reporting and documentation processes to improve AI systems' resource performance" (Article 40), and general-purpose AI documentation must state known or estimated energy consumption (Annex XI) [4]. Yet none of these instruments, and none of the thirteen catalogued initiatives, defines how the *quality* of an environmental claim should condition a deployment decision.

Recent governance research gates on magnitude. Carbon-aware governance gates embed carbon budgets in governance layers, recording provenance only in audit logs [5]. Sustainability-pyramid frameworks gate on readiness levels, arguing that launch blocking without carbon sign-off creates financial-grade controls [6]. Risk-management integrations treat AI carbon as a risk dimension in banking frameworks [7]. Meanwhile, measurement research has established that reliability varies

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sharply with disclosure: metadata-completeness tiers across 5,227 public models span roughly $1.2\times$ to $2\times$ estimation error [8], and spend-based fallbacks in tiered Scope 3 accounting overestimate by 10–40$\times$ [9]. Adjacent work has begun to act on estimate trust, through conformal calibration inside a carbon budget gate [10] and confidence-gated filtering of industrial emissions predictions [11], and evidence-driven release gates are established for functional quality [12]. The two literatures have not been connected: a system can sit comfortably under its carbon budget while its carbon number is untrustworthy, and every published gate would pass it.

Our claim is narrow and, to our knowledge, unheld: environmental *evidence quality*, not only impact magnitude, should gate AI deployment, operationalized as a provenance-derived confidence score crossed with an impact tier. The intellectual ancestry is the data-quality tradition in life-cycle assessment, where pedigree matrices and data quality indicators grade the provenance of an inventory figure [13], and the NUSAP uncertainty pedigree [14]. Recent work recommends indicator-style disclosure for AI-assisted product footprinting but proposes no score, no thresholds, and no gate [15]. We invert that tradition: where it assesses AI that produces carbon numbers, we score carbon numbers to gate the AI.

2 Mechanism

Provenance classes and confidence. Every assessment declares its measurement source, normalized to one of five provenance classes with bounded confidence ranges: *measured* (hardware telemetry; 0.85–1.00), *tool_estimated* (software power trackers; 0.60–0.85), *vendor_reported* (cloud-provider figures; 0.40–0.60), *manual* (analytic FLOP estimates; 0.20–0.40), and *unknown* (0.0). A caller-supplied score is honoured but never allowed to exceed its provenance cap: a claimed 0.95 on vendor-reported data becomes 0.60, and unknown provenance forces 0.0 regardless of any claimed score. Provenance is categorical and reported separately from quantified uncertainty, so a precise-looking number cannot launder a weak source.

Impact tiers. Metrics map to low, medium, or high impact per lifecycle phase (training, fine-tuning, inference, and LLM inference), using location-based carbon for tiering so market instruments cannot lower a gate outcome. Thresholds are explicitly provisional and are recalibrated from the measurement study (Section 4); an intensity-versus-baseline ratio can raise a tier at any confidence but lower it only with measured-grade confidence.

The decision matrix. Confidence bands (measured ≥ 0.70; estimated 0.35–0.69; unknown < 0.35) cross with impact tiers (Table 1). Confidence 0.0 short-circuits to *no_go* before matrix lookup: an undisclosed claim never passes. A *conditional_go* cannot be approved until it carries a dated, owned mitigation or an explicit exception with owner, expiry, and rationale, which makes reviewer overrides auditable rather than informal. Six controls (ENV-1 to ENV-6: boundary, energy accounting, dual location and market carbon, per-functional-unit efficiency, evidence confidence with disclosed uncertainty, and mitigation or exception) define coverage. Disclosed offsets are recorded but never subtracted from emissions and never improve any gate output.

Table 1: The impact by evidence-confidence decision matrix. Confidence exactly 0.0, meaning undisclosed provenance, returns *no_go* for every tier before matrix lookup.

Impact	measured	estimated	unknown
low	go	go	conditional_go
medium	conditional_go	conditional_go	no_go
high	conditional_go	no_go	no_go

3 Implementation and consistency evaluation

EVIDENCEGATE ships inside an open-source AI governance platform, not as a standalone proposal. The decision engine is a framework-agnostic Python package with 57 unit tests covering every matrix cell, band boundary, provenance cap, and the undisclosed-evidence short-circuit. Assessments persist as versioned governance evidence, and a release gate hooks the platform's existing deployment-approval workflow, returning HTTP 409 with the blocking reason when the latest assessment is *no_go* or an unmitigated *conditional_go* (15 integration tests). Reviewers get an interface showing the matrix position, control coverage, mitigation tracker, and version trail, and disclosures export in a format shaped to the European sustainability reporting standard for climate change [16]. Given the same assessment payload the gate decision is deterministic and reproducible.

Fixture consistency check. On 14 hand-labeled cases spanning all matrix cells, mitigation states, exception paths, and offset disclosures, the full mechanism reproduces 14/14 expected labels. Ablations degrade sharply (Table 2). We state the boundary plainly: the authors wrote both the fixtures and the engine, so this is a consistency and ablation illustration, evidence that the confidence dimension is load-bearing within the mechanism, and not an empirical validation. That is what the two studies below are for.

Table 2: Hand-labeled fixture ablation over 14 cases. A consistency check on author-written fixtures, not an empirical validation; reported to show that the evidence-confidence dimension is load-bearing.

Gate variant	Exact matches	Accuracy
Full mechanism	14/14	100.0%
No exception path	13/14	92.9%
Offset-credit gate	13/14	92.9%
No mitigation-review gate	11/14	78.6%
Carbon-only gate	7/14	50.0%
Generic sustainability score	6/14	42.9%
No environmental gate	3/14	21.4%

4 Registered validation studies

Measurement study. Four workloads (classical ML training, transformer fine-tune, retrieval-augmented inference, and batch embeddings) across five iterations, with each physical run measured three ways: a software power tracker (*tool_estimated*), operator-filled cloud billing (*vendor_reported*), and a documented analytic FLOP estimate (*manual*). The harness never fabricates a measurement; missing dependencies or billing entries produce rows explicitly marked unavailable. Outputs feed method-agreement statistics, pairwise MAPE and Bland-Altman limits of agreement, that recalibrate the provisional thresholds and confidence ranges. The design isolates the central question: the same run, at identical true impact, carries different provenance under each method, so we can report how often the gate decision flips on provenance alone, a quantity no magnitude-only gate can express.

Expert panel. Twelve synthetic system profiles cover all nine matrix cells, with the boundary cells doubled, and each profile is validated against the production engine so its expected labels are the engine's own output. Fifteen raters, five each from governance/risk/compliance, ML engineering, and sustainability roles, rate eight profiles apiece under a counterbalanced assignment giving ten ratings per profile with at least three per role. Two arms: baseline, showing raw measurements only, and module, showing the full EVIDENCEGATE output. Pre-registered hypotheses are H1, that displayed confidence correlates negatively with the recommendation ordinal (Spearman); H2, that the module arm raises ENV control coverage against a per-profile gold checklist (Wilcoxon signed-rank on paired coverage rates); and H3, that the matrix improves inter-rater agreement (mean

pairwise Cohen's κ and Fleiss' κ with bootstrapped intervals, target $\kappa > 0.60$). The analysis pipeline is implemented with exact statistics, environment-independent bootstrap intervals, and hand-verified test vectors, so results are computable the day collection closes.

5 Pathway to climate impact

Environmental reporting for AI is moving from voluntary to mandatory, but mandatory reporting only reduces emissions if the reported numbers are trustworthy enough to act on. Today a deployment decision can be signed off against a carbon figure with no disclosed provenance, and every published governance gate would permit it; the incentive to invest in real measurement is correspondingly weak. Making evidence quality a release criterion changes that incentive directly: a team that cannot substantiate its number cannot ship unconditionally, so measurement effort becomes the cheapest path to approval rather than an unrewarded cost. This is a leverage point rather than a direct abatement: it does not itself cut emissions, it makes the reductions others report verifiable and therefore actionable.

The near-term channel is regulatory. Article 40's harmonised standards for resource-performance reporting do not yet exist, and the standards bodies' current deliverables contain no environmental item [4, 17]. A provenance-scored, platform-integrated gate is a concrete and testable proposal for what those standards could require: anticipatory infrastructure rather than commentary. The mechanism is deployed in an open-source platform, the fixtures, profiles, harness, and analysis code are public, and both studies are designed for independent replication. We ask the workshop for critique of the two study designs, particularly the panel's baseline-arm construction and the defensibility of the provenance caps, while changes are still cheap.

Limitations. The fixture evaluation is author-constructed and non-empirical by design. The measurement study is single-site in its first run. The panel targets fifteen raters, sized for feasibility rather than power against small effects, and impact-tier thresholds remain provisional until recalibrated.

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